

MasteryFlow_V2 - Project Source Code

--- README.md ---

MasteryFlow V2

Upgraded adaptive learning prototype.

New features:

- Multi-concept question bank
- Basic / Intermediate / Advanced / Challenge difficulty
- Logical, numerical, conceptual and application questions
- Mastery-driven question selection
- Explainable feedback
- Teacher dashboard and overrides
- Learner path simulation
- Prerequisite knowledge graph

Run:

```
pip install -r requirements.txt
```

```
streamlit run app.py
```

--- requirements.txt ---

```
streamlit>=1.36
```

```
pandas>=2.0
```

--- app.py ---

```
import streamlit as st
```

```
from datetime import datetime
```

```
st.set_page_config(page_title="MasteryFlow V2", page_icon="?", layout="wide")
```

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```
PREREQS={"Fractions":[],"Ratios":["Fractions"],"Percentages":["Fractions"],  
"Linear Equations":["Ratios","Percentages"],"Graphs":["Linear Equations"],  
"Quadratic Equations":["Linear Equations"],"Probability":["Fractions","Percentages"],  
"Statistics":["Percentages","Probability"]}  
  
Q=[  
  
("F1","Fractions",1,"MCQ","Which is larger:  $\frac{3}{4}$  or  $\frac{2}{3}$ ?",[" $\frac{3}{4}$ ", " $\frac{2}{3}$ ", "Equal", "Cannot  
determine"], " $\frac{3}{4}$ ", " $\frac{3}{4} = 0.75$  and  $\frac{2}{3} \approx 0.667$ ."),  
  
("F2","Fractions",3,"Application","A tank is  $\frac{3}{5}$  full. If  $\frac{1}{10}$  is used, what  
remains?",[" $\frac{1}{2}$ ", " $\frac{2}{5}$ ", " $\frac{3}{10}$ ", " $\frac{7}{10}$ "], " $\frac{1}{2}$ ", " $\frac{3}{5} - \frac{1}{10} = \frac{1}{2}$ ."),  
  
("R1","Ratios",1,"MCQ","A recipe uses 2 cups flour for 3 cups water. Ratio  
flour:water?",["2:3", "3:2", "2:5", "5:2"], "2:3", "The quantities are 2 to 3."),  
  
("R2","Ratios",3,"Application","Boys:girls = 3:5. If there are 40 students, how many are  
girls?",["15", "20", "25", "30"], "25", "There are 8 parts;  $40/8=5$ , so girls=25."),  
  
("P1","Percentages",1,"Numerical","What is 20% of 150?",["20", "25", "30", "35"], "30", " $0.20 \times 150 = 30$ ."),  
  
("P2","Percentages",4,"Logical","A price rises 20% then falls 20%. Final  
price?",["Same", "4% lower", "4% higher", "Cannot determine"], "4% lower", " $1.20 \times 0.80 = 0.96$ ."),  
  
("L1","Linear Equations",1,"Numerical","Solve:  $2x + 6 = 14$ .",["2", "3", "4", "5"], "4", " $2x=8$ , so  $x=4$ ."),  
  
("L2","Linear Equations",3,"Application","Taxi: ?50 fixed + ?12/km. Bill ?170.  
Distance?",["8", "10", "12", "14"], "10", " $50+12k=170$ , so  $k=10$ ."),  
  
("G1","Graphs",2,"Conceptual","In  $y = 2x + 3$ , what is the  
slope?",["2", "3", "5", " $\frac{1}{2}$ "], "2", "In  $y=mx+c$ , m is slope."),  
  
("G2","Graphs",4,"Logical","Lines with slopes 2 and  $-\frac{1}{2}$   
are:",["Parallel", "Perpendicular", "Identical", "No relation"], "Perpendicular", "Their  
slope product is -1."),  
  
("Q1","Quadratic Equations",2,"Conceptual","Degree of  $x^2 + 3x + 1$ ?",["1", "2", "3", "4"], "2", "Highest exponent is 2."),  
  
("Q2","Quadratic Equations",4,"Logical","Roots of  $x^2 - 5x + 6 = 0$ ?",["1 and 6", "2 and  
3", "-2 and -3", "0 and 6"], "2 and 3", " $(x-2)(x-3)=0$ ."),
```

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```
("PR1","Probability",1,"MCQ","Fair coin: probability of heads?",["0","1/4","1/2","1"],"1/2","One favourable outcome out of two."),

("PR2","Probability",3,"Logical","Bag has 3 red and 2 blue. Probability of not blue?",["2/5","3/5","1/2","1/5"],"3/5","Not blue means red: 3/5."),

("S1","Statistics",1,"Numerical","Mean of 2, 4 and 6?",["3","4","5","6"],"4","(2+4+6)/3=4."),

("S2","Statistics",4,"Logical","Which has larger mean: [2,4,6] or [2,4,6,100]?",["First","Second","Equal","Cannot determine"],"Second","The large 100 raises the second mean.")
```

```
]
```

```
def init():
```

```
    if "learners" not in st.session_state:
```

```
        st.session_state.learners={
```

```
            "Aarav":{"m":{c:0 for c in PREREQS},"a":[],"o":[]},
```

```
            "Diya":{"m":{c:0 for c in PREREQS},"a":[],"o":[]}}
```

```
st.session_state.learners["Aarav"]["m"].update({"Fractions":.92,"Ratios":.82,"Percentage s":.76,"Linear Equations":.68})
```

```
st.session_state.learners["Diya"]["m"].update({"Fractions":.45,"Ratios":.38,"Percentages ":.30,"Linear Equations":.22})
```

```
def action(d,c):
```

```
    weak=[p for p in PREREQS[c] if d["m"][p]<.55]
```

```
    hints=sum(x["hint"] for x in d["a"] if x["c"]==c)[-5:] if False else sum(x["hint"] for x in d["a"] if x["c"]==c)
```

```
    m=d["m"][c]
```

```
        if weak:return "Remediate Prerequisite",f"{weak[0]} mastery is only {d['m'][weak[0]]*100:.0f}%."
```

```
    if m<.45:return "Practice",f"Mastery is {m*100:.0f}%; more evidence is needed."
```

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```
    if m<.70 or hints>=2:return "Review",f"Mastery is {m*100:.0f}% with {hints}
hint(s)."
```

```
    if m<.85:return "Challenge",f"Mastery is {m*100:.0f}%; test deeper transfer."

    return "Advance",f"Mastery is {m*100:.0f}% with prerequisites strong."
```

```
def target(m):

    return 1 if m<.45 else 2 if m<.65 else 3 if m<.82 else 4
```

```
def pick(d,c):

    used={x["id"] for x in d["a"]}

    pool=[x for x in Q if x[1]==c and x[0] not in used] or [x for x in Q if x[1]==c]

    return sorted(pool,key=lambda x:abs(x[2]-target(d["m"][c])))[0]
```

```
def update(old,ok,conf,diff,hint,time):

    e=(.65 if ok else 0)+conf/5*.15+diff/4*.10+(0 if hint else .10)

    if time>90:e-=.05

    return round(.75*old+.25*max(0,min(1,e)),3)
```

```
init()

names=list(st.session_state.learners)

name=st.sidebar.selectbox("Learner",names)

d=st.session_state.learners[name]
```

```
st.title("? MasteryFlow V2")

st.caption("Explainable Adaptive Learning & Intervention Engine")
```

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```
t1,t2,t3,t4=st.tabs(["? Adaptive Practice","? Teacher Dashboard","? Path Simulation","? Knowledge Graph"])

with t1:

    c=st.selectbox("Concept", list(PREREQS))

    q=pick(d,c)

    st.write(f"**Target difficulty:** Level {target(d['m'][c])}")

    st.caption(f"{q[3]} · selected difficulty {q[2]}")

    st.info(q[4])

    ans=st.radio("Choose your answer",q[5],index=None)

    conf=st.slider("Confidence",1,5,3)

    hint=st.checkbox("Used a hint")

    secs=st.number_input("Response time (seconds)",1,600,35)

    if st.button("Submit & Adapt",type="primary"):

        if ans is None: st.error("Choose an answer first.")

        else:

            ok=ans==q[6]; old=d["m"][c]; new=update(old,ok,conf,q[2],hint,secs)

            d["m"][c]=new;

            d["a"].append({"id":q[0],"c":c,"correct":ok,"hint":int(hint),"difficulty":q[2],"time":datetime.now().isoformat()})

            ac,why=action(d,c)

            st.success("Correct!" if ok else f"Correct answer: {q[6]}")

            st.write("**Explanation:**",q[7])

            st.metric("Updated mastery",f"{new*100:.0f}%")

            st.warning(f"**Next action: {ac}** ? {why}")

            if ac in ["Advance","Challenge"]: st.success("? Mastery is high enough to increase cognitive difficulty.")

    if d["a"]:
```

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```
st.subheader("Recent evidence")

st.dataframe(d["a"][-6:],width="stretch")

with t2:

    rows=[]

    for n,x in st.session_state.learners.items():

        for c,m in x["m"].items():

            ac,why=action(x,c);

    rows.append({"Learner":n,"Concept":c,"Mastery":f"{m*100:.0f}%", "Next
    Action":ac, "Reason":why})

    st.dataframe(rows,width="stretch")

    st.subheader("Teacher Override")

            tn=st.selectbox("Learner",names,key="tn");
tc=st.selectbox("Concept", list(PREREQS),key="tc")

            ta=st.selectbox("Override",["Advance","Practice","Review","Remediate
    Prerequisite","Challenge","Teacher Intervention"])

    note=st.text_input("Teacher note")

    if st.button("Save override"):

st.session_state.learners[tn]["o"].append({"concept":tc,"action":ta,"note":note})

    st.success("Override logged.")

with t3:

    st.subheader("Same curriculum, different paths")

    for n in names:

        st.write(f"### {n}")

        cols=st.columns(4)

        for i,c in enumerate(list(PREREQS)[:4]):
```

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```
with cols[i]:

    st.metric(c,f"{st.session_state.learners[n]['m'][c]*100:.0f}%")

    st.caption(action(st.session_state.learners[n],c)[0])

    st.info("Strong evidence increases difficulty; prerequisite gaps trigger remediation.")

with t4:

    st.subheader("Prerequisite Knowledge Graph")

    for c,p in PREREQS.items():

        st.write(f"***{'', '.join(p) if p else 'Foundation'} ? {c}***")

    st.caption("The engine checks prerequisite mastery before advancement.")

st.divider()

st.caption("V2 extension: adaptive question bank, difficulty progression, logical/application questions, explanations and teacher controls.")
```